AthenaStudies

History of Science Lecture and Book Notes

Vrije Universiteit
History of Science
Notes

1.1 Part One: Before the Computer

1.1.1 Chapter 1 - When Computers Were People

Computer was occupation up to WWII

1.1.1.1 Logarithms and Mathematical Tables

Sixteenth and seventeenth centuries - information processing on a large scale using human computers was for the production of mathematical tables
Then eighteenth century specialized tables like Nevil Maskelyne, to produce navigational tables, Nautical Almanac.

1.1.1.2 Charles Babbage and Table Making

Charles Babbage interested in the problem of table making and the elimination of errors in tables.
Circa 1790 great French table-making project organized by Baron Gaspard de Prony.
De Prony took division of labor, which De Prony organized his table-making “factory” into three sections. mathematicians, middle management, human computers
Babbage inspo from de Prony’s invent a ma- chine for making tables.
Babbage called his machine a Difference Engine
Was government sponsored 1820s
Mechanical technology not there
1833, Babbage prototype Difference Engine that was too small for real table making and lacked a printing unit
New idea: Analytical Engine. - it would be capable of performing any calculation

1.1.1.3 Clearing Houses and Telegraphs

1820 no office machinery or large scale clergy
one large-scale data-processing organiza- tion This was the Bankers’ Clearing House in the City of London,
The Bankers’ Clearing House was an organization that processed the rapidly in- creasing number of checks being used in commerce.
telegraph, which began to compete with the ordinary postal system in the 1860s.
it began as a solution to a communica- tions problem in the early rail system.
In 1870, the British government stepped in to integrate the systems into a national telegraph network.

1.1.1.4 Herman Hollerth and the 1890 Census

Compared to Europe, the United States was a latecomer to large-scale data process- ing. When Britain, Germany, and France were industrializing in the 1830s, the United States was still primarily an agricultural country.
Bureau of the Census in Washington, DC. : The census of 1880 - tally system
census problem solved by Herman Hollerith (1859–1929). He later devel- oped a mechanical system for census data processing, commercialized his invention by establishing the Tabulating Machine Company in 1896,
the Hollerith Electric Tabulating System adopted for the 1890 census.

1.1.1.5 AMERICA'S LOVE AFFAIR WITH OFFICE MACHINERY

1880s, 1890s , office equipment, American phenomenon. not be seen in Europe until the new century or after World War I.
There were two main reasons for America's affinity for office machinery. First, because of the American office's late start compared with Europe, it did not carry the albatross of old-fashioned offices and entrenched archaic working methods
Example : British Prudential Assurance Company vs American
America was gadget-happy, perceived as modern
Systematizers set about restructuring the office, introducing typewriters etc.

1.1.2 Chapter 2 : The Mechanical Office

IN 1928 THE WORLD'S top four office-machine suppliers were Remington Rand, National Cash Register (NCR), the Burroughs Adding Machine Company, then—IBM,
Remington Rand - typewriters and filing systems for information storage.
Burroughs dominated the market for adding ma- chines used for simple calculations.
IBM dominated the market for punched-card accounting machines.
And NCR, -cash registers in the 1880s, also became a major supplier of accounting machines.

1.1.2.1 The Typewriter

typewriter that ushered female workers into the office. prior "copyist."
1867, Christopher Latham Sholes,invented machine that became the first Remington type-writer. Had speed was due to a "keyboard and type-basket" arrangement,
QWERTY keyboard layout to prevent jams
The first ones were sold in 1874 then a big boom
Beginning in the 1880s, organizations came into being to train young work- ers (mostly women) for the new occupation of "type-writer," later known as a "typ- ist."
key features of the office- machine industry: the perfection of the product and low-cost manufacture, a sales organization, and a training organization for the technology.

1.1.2.2 The Rands

James Rand Sr - he invented index system 1898 Rand organized the Rand Ledger Company,
Rand jr, made Kardex System had his own company for a bit then re merged with father
In 1927, the company merged with the Remington Type- writer Company and became known as Remington Rand.

1.1.2.3 The First Adding Machines

Adding machines 4 processing information. The first commercially produced adding machine was the Arithmometer - not reliable
Sturdy, reliable Arithmometers became available in the 1880s, its speed of operation was too slow for routine adding in offices
The solvers of these two critical problems, Dorr E. Felt and William S. Burroughs, they created, the Comptometer and the Burroughs Adding Machine, "key-driven.".
As with the typewriter, the skill to use the Comptometer was not easily acquired. -> training schools
Burroughs began to develop an adding machine for banks
In 1885 Burroughs founded the American Arithmometer Company, build his "adder-lister."

joined by dozens of other calculating-machine manufacturers such as Dalton (1902), Burroughs supplied not only adding machines but also the in- tangible know-how of incorporating them into an existing business organization = survival

1.1.2.4 The National Cash Register Company

innovative forms of selling, by the National Cash Register Company in the 1890s.
NCR, John H. Patter- son, the company's founder,
The first practical cash register was "Ritty's Incorruptible Cashier" in 1879.
1888 James Ritty established a small "inven- tions department" for this purpose. first formal R&D
After Patterson death major NCR redirection,Class 2000 accounting machine
NCR's key sales practices of the industry. NCR Primer. salesman training school

1.1.2.5 Thomas Watson and the Founding of IBM

1895 Watson with NCR Patterson fired him.
1896, Hollerith incorporated his business as the Tabulating Machine Company (TMC).
An offer to take over TMC Charles R. Flint, - 1911
He proposed merging the Tabulating Machine Company with two other firms—the Computing Scale Company, and the International Time Recording Company, = Computing-Tabulating-Record- ing Company, or C-T-R.
C-T-R become IBM in 1924,
ultimate success of IBM: sales organization based on that of NCR, the "rent-and-refill", and technical innovation. Made the Hundred Point Club IBM virtually recession-proof.
The second source of IBM's financial stability was its punched-card sales
Technical innovation response to competition the Powers Accounting Machine Company, which was acquired by Remington Rand,
In the early 1930s IBM leapfrogged the competition by announcing the 400 Series accounting machines

1.1.3 Lecture 1: Prehistory of Computing

Machines and technology

- No trust of technology 18 and 19 century Example: "Chess Playing Turk" entertainment circus attractions, tricks/illusions, technology was there to fool
- Pascaline - calculating devices remained relatively unknown and not accessible to masses in Paris 19th century
- Late 19th century cash register comes about, one of the first exposure to general audience, still took some years until acceptance
- Babbage all about efficiency in data processing
- Frederik Taylor shop management and the principles of scientific management early 1900s company design and workflow, scientific management
- Example Ford 1913 applies Taylor's methods
- Henry Fayol french dude who worked also with industrial management and efficiency
- Trust in numbers was a rarity as well, not trivial
- Trust in number grows: the 1890 US Census uses the Hollerith machines and trusts in them.
- European continent was politically more left than the US and UK.
- Less value in statistics

Three traditions of how to automate which lead to computers

- Administration
- Process control
- Science

Administration

- Office 1876 handwritten , office in 1920 typists and women in the office, Modern places
- Had: punched card typing, calculators, typewriters, sorting machines, counting machines, tabulating machines
- Machines increased efficiency
- These machines integrated naturally in US in Europe it was introduced by government to aid industrialization
- Example: 1911 Amsterdam International Exhibition of Modern Office Equipment
- ROBAVER 1922 succeeded in implementing machines in business(Netherlands)

Process Control

- Huge companies like Shell with huge machines needed control over their factory processes , Used to be done manually then machines
- Outlines analogue to the process
- Analogue machinery example. DSM(1956)

Science and engineering

- Computers were people active in lots of things like weather prediction
- Econometry in Europe more than US

All had outlines

1.1.4 Chapter 3 : Babbage's Dream Comes True

1.1.4.1 Babbage's Analytical Engine

Babbage never built Difference Engine because new invention, the Analytical Engine,
the Analytical Engine embodies most of the important functions of the modern digital computer.
Analytical Engine the mill and the store and Jacquard loom - inspo
cause of Babbage's fail- ure was that he was pioneering in the 1820s and 1830s a digital approach to com- puting some before mechanical technologies sufficient

1.1.4.2 The Tide Predictor and Other Analog Computing Machines

physical model, or analog, of the system to be investigated.
Example: The most important analog computing technology 19th century, was the mechanical tide predictor.by Lord Kelvin invented a useful tide- predicting machine. -> Tide tables
Analogue —a specific problem required a specific machine.
Vannevar Bush (1890–1974) Bush's differential analyzer, built between 1928 and 1931, was not a general- purpose computer in the modern sense, but it addressed wide range of prob- lems

1.1.4.3 A Weather-Forecast Factory

One of the first advocates of the team computing approach was the Englishman Lewis Fry Richardson (1881–1953), who pioneered numerical meteorology— his book, which was published in 1922. In it he de- scribed a global weather-forecast factory.

1.1.4.4 The Scientific Computing Service

Leslie John Comrie (1893– 1950), who would transform the approach to computing both in Britain and the United States.
Comrie's great insight was to realize. For most calculations, he found that his "calculating girls" equipped with ordinary commercial calculating machines did the job per- fectly well.
Perhaps Comrie's major achievement at the Nautical Almanac Office was to bring punched-card machines—and therefore, indirectly, IBM—into the world of scientific computing.
In 1937 he took the unprecedented step of setting up in business as a commercial operation, the Scientific Computing Service Limited. It was the world's first for-profit calcu- lating agency.

1.1.4.5 The Harvard Mark 1

in the period 1935–1945, at least ten machines constructed during this period, such as AT&T
IBM Automatic Sequence Controlled Calculator. = Harvard Mark I, 1937–1943.
Hathaway Aiken, proposal to construct a large-scale digital calcu- lator. Aiken inspired by Babbage
first fully automatic machine to be completed
the machine was programmed by a length of paper tape, "operation codes".
No "con- ditional branch"= less impressive than Babbage's Analytical Engine,
The IBM Automatic Sequence Controlled Calculator was officially dedicated at Harvard on 7 August 1944, and Aiken was an egoistic self serving credit-hogging dick

1.1.4.6 Theoretical Developments and Alan Turing

Alan Mathison Turing and the American mathematician Alonzo Church. Both were trying to answer a question: the Entscheidungsproblem
Turing and Church came up with equivalent results in the mid-1930s.
Tur- ing used a conceptual computer, later known as the Turing Machine.
It was computing reduced to its simplest form and "universal"
the Turing Machine embodied all the logical capabilities of a modern computer.
John Von Neumann, Turing's work significantly influenced von Neumann and, hence, the invention of the modern electronic computer.

1.2 Part 2: Creating the Computer

1.2.1 Chapter 4 : Inventing the Computer

need for mathematical computation from war
The scientific war effort in the United States was administered by the Office of Scientific Research and Development (OSRD).

1.2.1.1 The Moore School

computer for ballistics computations BRL and the Moore School of Electrical Engineering to- gether
By early 1942, the Moore School basement of the building, the differential analyzer—affectionately known as "Annie"—
No quick and effective calculating tech means harder to use newly developed weapons

1.2.1.2 The Atansoff-Berry Computer

1942 John Mauchly proposed electronic computer to eliminate this bottleneck.
Iowa State University project was begun in 1937 by John Vincent Atanasoff, with Clifford Berry, and by 1939 they went on to build a complete machine = Atanasoff-Berry Computer (ABC). with binary arithmetic and electronic switching elements.
Mauchly visited Atanasoff in June 1941 did he copy? unclear

1.2.1.3 Eckert and Mauchly

John Presper Eckert. worked together with Mauchly
Eckert worked on delay-line storage system for a Moving Target Indicator (MTI) (radar systems)

1.2.1.4 ENIAC and EDVAC: The Stored-Program Concept

“Project PX” to construct this ma- chine, known as the ENIAC (Electronic Numerical Integrator and Computer).
concerned the large number of vacuum tubes the machine would contain. Eckert’s idea: voltage was reduced life was extended of tubes.
The ENIAC was Eckert and Mauchly’s project.
Problem: reprogramming solution: wired-up for a specific problem
In short, there were three major shortcomings of the ENIAC: too little stor- age, too many tubes, and too lengthy reprogramming.
John Von Neumann became a consultant to the ENIAC group = new design, “stored-program computer,” = ENIAC’s successor: the EDVAC
Problem: storage limitations solution: Eckert proposed the use of a delay-line storage
EDVAC stored program = would enable a program to treat its own instructions as data.
Eckert and Mauchly’s contributions = delay-line research, von Neumann, Herman Goldstine, = mathematical-logical structure of the machine.
Another key decision was to use binary to represent numbers

1945, entitled A First Draft of a Report on the EDVAC, Von Neumann’s sole authorship
von Neumann and Goldstine, idea move into the public domain as rapidly as possible,
for Eckert and Mauchly, computer = entrepreneurial business opportunity

1.2.1.4.1 The Engineers vs. the Logicians

Eckert was particularly irked, credit on von Neumann, but they also allowed him to promote his abstract-logical view of com- puters, vs Eckert’s down-to-earth engineering viewpoint.
his academic interests conflicted with Eckert’s commercial interests. He noted that the government had paid for the de- velopment of the ENIAC and the EDVAC and that therefore it was only fair that these ideas be placed in the public domain.
In November 1945, the ENIAC at last sprang into life.
In March 1946 in Philadelphia Eckert and Mauchly set up in business as the Electronic Control Company to build computers.

1.2.1.4.2 The Moore School Lectures

the knowledge of stored-program computing was effectively transmitted to the outside world. He decided to organize a summer school for thirty to forty invitation-only participants, one or at most two representatives from each of the major research or- ganizations involved with computing - The Moore School Lectures,

1.2.1.5 **Maurice Wilkes and Edsac**

the first two computers to be completed were both in England, at Manchester and Cambridge universities..
A mechanical com- puter—specified by Alan Turing—was operational in 1940, and this was followed by an electronic machine, the Colossus, in 1943.
The Manchester computer was the first to become operational. By Max Newman
F. C. (later Sir Frederic) Williams: key problem in developing a computer was the mem- ory technology. he developed a simple memory system based around a commercially available cathode-ray tube.
Williams and his assistant built a tiny computer to test out the new memory system. 1948, and the “Manchester Baby Machine” estab- lished incontrovertibly the feasibility of the stored-program computer.
Maurice Wilkes at Cambridge University. Invited to Moore School Lectures He arrived with only two weeks left of course and saw both the Harvard Mark I
Upon his return to Cambridge in October 1946, focus in pro- gramming and mathematical applications—
the machine the Electronic Delay Storage Automatic Calculator; the acronym EDSAC
1949 The world’s first practical stored-program computer had come to life, and with it the dawn of the computer age.

1.2.2 **Lecture 2: Creating Computers**

Z1 1938 Konrad Zuse in Germany computer research program, No one was interested
Yamashita type statistical machine, 1944-1948 working in 1946 in Tokyo Japan with a keyboard entering machines very atypical
Japan was using IBM machines for tabular exercises before the war and during the war when they had no access they made their own
After WWII need for scientific calculations exploded. Example: NASA
In Europe

- Rebuilding in the 1940s with Marshall Plan
- Mix of fatalism and optimism

1946 Moore School - Mathematisch Centrum 1946 a rise in the trust in numbers and math in Netherlands so they created schools and funding
Hans Freudenthal(1905-1990) - Rekenmachines winnen den oorlog(calculators win the war) 1946
Society could and should improve
Harvard Mark III Time Magazine January 1950
Wanting to be part of modernity
Europe:
Love hate relationship with the US. wanting to be better and different than US
Jean Jacques Servan-Schreiber(1924-2006) exemplifies “The American Challenge” ie keeping up with and bettering tech, modernity
American culture overwhelming europe was seen as serious problem
December 1948 “Manchester Baby” war trodden Europe was ready for the Computer Age as well as Japan
Dutch Dinosaurs:

- Land reclamation and waterworks, required huge calculations effort in 1910s 1920s by Lorentz and Lely

- Tradition of process control: watch the deijks Johan van Veen, Rijkswaterstat did calculations on water and dam movements and canals. Used machines analogue computers
- 1952 ARRA (a dutch dinosaur in Amsterdam) put to work by the minister first dutch computer a one time working random number generator
- ARRA II 1954 transistors no tubes and really a whole new computer
  Electrologica, spin off 1956 took the ARRA II and named it X1 and sold it
- 1950 first job performed by ARRA but then in the early 1960s 179 jobs were performed on the X1
- Until late 1950s early 1960s human computers then computers

Delft Dinosaur

- Willem van der Poel built the ARCO in 1950 (Automatic Relais Calculator for optical problems)
- Programs hard-wired but interchangeable
- Very slow but used
- ZERO built in 1951 for Dutch telephone company facilitating the telephone connections instead of operators
- 1955 Van der Poel designed ZEBRA for the standard Telephone and Cable Company(STANTEC) 1957

Eindhoven

- PETER not very reliable helps to test machinery

IBM and Phillips have an agreement, IBM buys Phillips parts and Phillips doesnt go into computer market

PASCAL and STEVIN most notable followups on PETER lightening work load of Phillips engineers

Shell in 1953 made newspapers with their machine MIRACLE

How much work did the machines do? Idk probably not much but portray modernity to public

History of AI

- AI used today is really from the 2000s
- But there are predecessors
- Idealistic researchers, very socially minded wanting to better the world
- 1956 Dartmouth Conference: The Founding Fathers of AI (mostly mathematicians)
- Kurt Goedel, incompleteness theory(1931)
- Alonzo Church, lambda calculus(1936)
- Alan Turing, a-machine(1936)
- Investigation into the working of the human brain
- In decision making(economics, administration, process control)
- How do we cope with language
- A.M. Turing “Intelligent Machinery” (1948) founding a field of interest
- Cybernetics- psychology, mathematics, biologists etc

Modern Times (1936)

- American reception of Taylorism - modernity
- Taylorism, also known as scientific management, is a theory of management that analyzed how workers should be motivated.

European tradition of dystopian literature

- Karel Capek, R. U. R. (1920) robot in media and human-like
- George Orwell, 1984 in (1948) robots in media and totalitarianism with tech

Public not very involved but occasionally informed then in 40s and 50s increase of general science info

Rise of computing technology as part of modernity pinpointed to the 50s

This lead to the Golden Age of Science fiction like I, Robot (1950)

1.2.3 Chapter 5 : The Computer Becomes a Business Machine

computer to electronic data-processing machine rather than a mathe- matical instrument.
IBM had recognized this change, 1951, reoriented its R&D, manufacturing, and sales organizations
Britain even had the world's first computer on the market. Ferranti Mark I, based on the Manchester University computer and delivered in February 1951. Not sucessful
three types of firms that entered the computer industry: electronics and control equipment manufacturers(RCA,GE), office-machine companies(IBM and co.), and start-ups.
first entrepreneurial computer firms = Eckert and Mauchly's Electronic Control Company
There were other start ups most simply faded away, or were absorbed
When Eckert and Mauchly established their Electronic Control Company in March 1946, they saw potential for computers in business data processing, as 1946, the Bureau of the Census agreed to buy a computer from Eckert and Mauchly,
the computer was officially named UNIVAC, for UNIVersal Automatic Computer. $$$$
UNIVAC was the use of magnetic-tape stor- age to replace the millions of punched cards
Magnetic tape chemistry had to be specially developed
magnetic-tape equipment for the UNIVAC was both the most ambitious and the least successful
Next for Northrop Aircraft Corporation made the BINAC (for BINary Automatic Computer).
The BINAC smaller; it would be a scientific machine, not a data processor; no magnetic tape
By the end of 1947 Eckert and Mauchly's incorporated the company as the Eckert-Mauchly Computer Corporation (EMCC) need $$$

1.2.3.1 IBM:Evolution, Not Revolution

1940s IBM delayed them being turned into products b/c uncertainty of the market.
IBM respond to three challenges: product obsolescence, electronics, and the computer.
IBM = "evolution not revolution." Thomas Watson Sr., leader of IBM
August 1944 Howard Aiken refused to acknowledge IBM's role in Mark I
Revenge = a "supercalculator" t== Selective Sequence Electronic Calculator (SSEC).
the SSEC was a one-of-a-kind development, never became part of the mainstream of stored-program computing.
Later the IBM Model 604 provided "speed and flexibility of operation unmatched by any calculator in the market for some time." later the heart of a powerful computing setup CPC
The CPC was developed in 1947 in a cooperative effort between IBM and Northrop Aircraft,
The CPC was not a true stored-program computer, reliable calculating tool in 1949 and 1950
"calculator" instead of a "computer." Watson insisted on this terminology, referred to a human
IBM also had two further computers under development in 1949: the Magnetic Drum Calculator (MDC) and the Tape Processing Machine (TPM). The MDC was cheaper than UNIVAC and true stored-program computing, The key = magnetic drum for the main memory
IBM's Tape Processing Machine on par with the UNIVAC.
If IBM had pressed on with either or both of these computers in 1949 or 1950, it would have dominated the market for data-processing computers much earlier than it did.
The delay of the MDC was due largely to conservatism in IBM's marketing
The delay with the Tape Processing Machine, however, had quite different causes: the start of the Korean War and Development of the Defense Calculator,
Watson's decision to build the Defense Calculator in the summer of 1950, while accelerating IBM's entry into scientific computers, dashed its chances of an early entry into the data-

processing market. It was a business error, one that allowed Eckert and Mauchly to capture the early market for data-processing computers with the UNIVAC.

1.2.3.2 UNIVAC Comes to Life

the Eckert-Mauchly Computer Corporation became a wholly owned subsidiary of Remington Rand.
UNIVAC was being sold at perhaps a third of its true cost. First they approached the Census Bureau to try to renegotiate the price,
While these negotiations were going on in the background, Eckert and Mauchly focused on completing the first UNIVAC.
1951, UNIVAC was functioning as a computer and would soon be ready to commence its acceptance tests for the Census Bureau
Over the following year, two more UNIVACs were completed and delivered to the government,
In late 1952, Remington Rand achieved a spectacular publicity stunt for UNIVAC. The company persuaded the CBS television network to use UNIVAC to predict the outcome of the presidential election.
Before that date, while some people had heard about computers, very few had actually seen one; after it, the general public had been introduced to computers and had seen at least a mock-up of one. And that computer was called a UNIVAC, not an IBM.

1.2.3.3 IBM's Big Push

It was the launch of the Census Bureau UNIVAC galvanized IBM into action
IBM Models 701, 702, and 650.
The development cost of the three concurrent projects was enormous,
Surprisingly, many conservatives within IBM still argued against developing the Tape Processing Machine, maintaining as late as 1953 that conventional punched- card machines would be cheaper and more cost-effective. However, this failed to take account of the fact that computers were hot news
Cost- effectiveness was no longer the only, or even the most important, reason for a busi- ness to buy a computer.
The first Model 701 scientific computer 1952, and IBM derived the maximum publicity
Although the IBM 702 Electronic Data Processing Machine (EDPM) were major differences to UNIVAC that enabled IBM's product quickly to outdistance the UNIVAC in the marketplace.
Instead of the mercury delay lines chosen for the UNIVAC, IBM decided to license the Williams Tube technology developed for the Manchester University computer
Another advantage over the UNIVAC was that IBM's computers were size shape
IBM's greatest advantage was its reputation as a service-oriented vendor. Had courses for users and customer support and fixers
In fact, neither the 701 nor the 702 proved to be quite the "UNIVAC-beater" that IBM had hoped. For both machines, the Williams Tube memories proved troublesome, making them unreliable by IBM's normal high standards. By 1953, however, a new technology known as core memory had been identified.
IBM mounted a crash research program to develop core memory into a reliable product. More reliable and faster core-memory versions of the 701 and 702 were announced in 1954 as the Model 704 scientific computer and the Model 705 EDPM.
According to Watson junior, 1955 was the turning point for IBM. That year, orders for IBM's 700 series computers exceeded those for UNIVACs.
Yet it was not the large-scale 700 series that secured IBM's leadership of the industry but the low-cost Magnetic Drum Calculator.

1952 Remington Rand had acquired Engineering Research Associates, which developed UNIVAC 1103. In mid-1955 Remington Rand merged with Sperry Gyroscope—and was renamed Sperry Rand—to further develop its technology.
in 1954 or 1955 it had already begun to slip behind IBM. Part of the reason for this was a failure in Remington Rand's marketing:
other reasons for Remington Rand's decline were infighting

1.2.3.4 The Computer Race

In the early 1950s the electronics and control manufacturers participate in the computer business by making machines for science and engineering
companies faced with the same decision: exit the business or compete with IBM. Of the major firms, only three—RCA, GE, and Honeywell—chose to commit the resources to stay in the race.
by far the most vulnerable firms were the small computer start-ups, of which very few survived into the 1960s. the Control Data Corporation (CDC) startup that survived
Like IBM, the office-machine companies had prudent doubts about the market for business computers and initially did nothing more than put electronics into their existing products in an evolutionary way.
by the end of the 1950s the major players in the computer industry consisted of IBM and a: Sperry Rand, Burroughs, NCR, RCA, Honeywell, GE, and CDC. Soon, journalists would call them IBM and the seven dwarfs.
The engine that would power that growth was a new computer, the IBM 1401.

1.2.4 Chapter 6 : The Maturing of the Mainframe: The Rise of IBM

THE 1957 FILM Desk Set, was also the first major motion picture to feature an electronic computer.with IBM Corporation.

1.2.4.1 The IBM 1401

Designed by Eliot Noyes, a whole computer system. While competitors "CPUs," and tended to neglect the system as a whole.
The origin of the 1401 was the need to create a transistorized follow-up to the tube-based model 650 Magnetic Drum Computer + core memory
sound reasons why most of IBM's customers were still resisting the computer: cost, reliability questionable, problem of hiring programming staff to develop applications software, since IBM provided little programming support at this time.
IBM also had to find a technological solution to the programming problem.
The solution that IBM offered was a new programming system called Report Program Generator (RPG) it became one of the most used programming systems in the world
IBM developed entire program suites for the industries that it served
The IBM 1401 was announced in October 1959, and the first systems were delivered early in 1960
The 1401 excellent computer, reason 4 success = 1403 "chain" printer

1.2.4.2 IBM and the Seven Dwarfs

In 1963 UNIVAC started to break even at last. Yet the machine that brought profits, the UNIVAC 1004, was not a computer at all but a transistorized accounting machine
IBM's other business-machine rivals, Burroughs and NCR, their strategy was safety first: the banking and retail sectors, respectively.

the fastest-rising star in the computer scene of the early 1960s was Control Data Cor- poration, the start-up founded in 1957 by the entrepreneur William Norris. Control Data had evolved a successful strategy by manufacturing mainframes with a better price performance than those of IBM.
The only other important survivors in the mainframe industry of the early 1960s were the computer divisions of the electronics and control giants—RCA, Honeywell, and General Electric. All three had decided to make the necessary in- vestment to compete with IBM. for all of them the computer was not a core business but an attempt to enter a new and unfamiliar market.
Honeywell was planning to launch its model 200 computer that would be compatible with the RCA, to develop a range of IBM-compatible computers
General Electric had plans to announce a range of three computers—one small, one medium-sized, and one large—in late 1963.

1.2.4.3 Revolution, Not Evolution: System/360

IBM problem: mishmash of incompatible product lines
IBM was being run almost as a federation of small companies instead of an integrated whole.
established the SPREAD task group solution = New Product Line that was to consist of a range of compatible computers to replace all of IBM's existing computers.
was a big secret, The New Product Line was one of the largest civilian R&D projects ever undertaken. "IBM's $5 billion gamble" , dubbed System/360
December 1963 when Honeywell announced its model 200 computer.
later the entire System/360 product line was launched. And made a big splash to eclipse Honeywell
System/360 has been called "the computer that IBM made, that made IBM."

1.2.4.4 The Dwarfs Fight Back

Perhaps the most serious design flaw in System/360 was its failure to support time-sharing technology was secondary to marketing in IBM's success
to compete with IBM at all, the seven-dwarf competitors had to aim at one of IBM's technical weak spots.
The first and most confrontational response was to tackle IBM head-on by making a 360-compatible computer with superior price performance - RCA, announced its range as Spectra 70 in December 1964
Hence the second and less risky strategy to compete with System/360 was product differentiation—that is, to develop a range of computers that were software- compatible with one another but not compatible with System/360. This was what Honeywell did
The third strategy to compete with IBM was to aim for niche markets Control Data, for example, recognized no really big machines, made only giant number-crunching computers
General Electric recognized IBM's weakness in time-sharing,
Burroughs and NCR were able to build on their special relationship with bankers and retailers, first computer recession of 1970–1971.
Both RCA and General Electric decided to give up the struggle and terminate their in-the-red computer divisions.
IBM and the BUNCH—standing for Burroughs, UNIVAC, NCR, Control Data, and Honeywell.

1.2.4.5 Systems Men and Management Information Systems

1st application electronic computing administrative processes ex.payroll, "electronic data processing," or EDP

One of the most influential groups of professional computer enthusiasts were the so-called systems men. incarnation of the “systematizers”
The goal of systems men = develop systems of management that would eliminate waste and inefficiency at all levels of the organization.
In the late 1950s they introduced the management information system (MIS).
The central idea behind MIS was that computers and communications tech- nologies could be mobilized to create a comprehensive picture of the total information environment of a complex and distributed organization.
the concept of “automation.” John Diebold argued that computer control systems could be used to bring about a second industrial revolution in manufacturing.

1.2.4.6 The Long March of Computerization

In 1962, Ross Perot founded Electronic Data Systems (EDS), the first (of many) computer facilities firms, which would establish and operate for its clients an entire computer installation; other firms emerged to develop the applications software
One obvious candidate for computerization was the financial sector. ERMA (Electronic Recording Machine, Accounting).

1.2.4.7 The Computer Becomes a Commodity.

Some larger firms had already developed their own electronic data-processing divisions in-house; oth- ers relied on the growing number of third-party consulting firms,While the standardization around the System/360 architecture eliminated certain forms of competition, it in- troduced others.
Faced with compe- tition from lower-cost compatible competitors from companies like Siemens, the mainframe business suddenly became a lot less profitable.
By the mid-1970s, the mainframe had matured and become a commod- ity, while IBM’s expertise in systems integration was increasingly being taken over by software or third-party provider
And perhaps more signifi- cantly, the increasingly diverse set of industries and disciplines that had developed a taste for computing in the 1960s began to develop new modes of computing that IBM was much less capable of providing.
IBM’s every move was being scrutinized by the Antitrust Division of the Department of Justice.

1.2.5 Lecture 3

Sounds of software
Early machines/automated calculators
EDSAC by Maurice Wilkes, Cambridge (1949)

- ‘God save the queen’.

reversed engineered the sound
Something similar happened on the MIRACLE.
It hummed the Dutch national anthem.
Engineers were listening to the execution of the code that they fed to the machine.
Sounds were used for:
As a way of checking the machine

i.e was the machine doing what it was supposed to do → was it not stuck at a specific procedure/or doing something completely different than expected).

No visual confirmation (there were no screens, cmds, monitors).

As a way to figure out how a machine works/operates

Announcing the program had terminated

Auditive monitoring

Specific navigation within a program

Part of the instruction.

The book 'The preparation of programs for an electronic digital computer' by Wilkes, Wheeler and Gill (1951) was the first handbook about "preparing programs" and made people aware of the sounds produced by machines.

Transistorised machines didn't produce any sounds. They were silent.

connecting a speaker and having the current run through the speaker.

Transistorised machines that had speakers attached:

ARRA (1951)

ARRA II (1954) transistors: with speaker

Use of metaphor

Typically, metaphors are used as reassuring names for new things. This is also what happened to early machines.

There were also attempts from language theory to make a machine understand natural language:

1950s: translating programs attempted

ELIZA (1946) by Joseph Weizenbaum

Allowed communication between humans and machines (it gave responses to someone who was typing).

Scientists used metaphors to explain how computers worked.

They used the human brain as orientation.

Turing's man (book) by Bolter argues about the relativiness and consequences of metaphors.

Bolter (1984) argues that using metaphors is defining technology and he argues that metaphors come back to real life.

E.g., computer memory acting as human memory.

E.g., thinking used for the UNIVAC 120.

E.g., the ENIAC and HARVARD MARK 1 were presented as electronic brains.

technology was oriented on nature by creating metaphors of natural phenomena.

Examples of metaphors:

Program as a word originated w.r.t. the ENIAC (1946)

Originally, a program was a way of executing something. Later on, a program was used to prepare something.

Programming used as a verb (1948)

Memory w.r.t. mercury delay lines and capacitors

Memory is originally stored data, but the term was copied from nature.

AI

Animal and human behavior can be explained by automated and algorithmic processes.
In 1948, Turing is trying to create a machine that mimics the human brain.
Turing creates a machine that can play chess against a human opponent.
Partly, the metaphor AI had its origin in pre-war research.
Most notably, in research about the psychology of the brain.
Brain mechanisms and intelligence (1929) by Karl Lashley.
This research leads to 2 directions in research in brain psychology:
Psychometry;
And later cybernetics
Trying to simulate/emulate brain functions.

Other metaphors that existed for programming language, before its usage:
Plugging
Autocoding, autocoding systems

In 1960 the term became programming language and the compiler became a translator with the arrival of COBOL, FORTRAN and ALGOL.
For logicians, it was a language..

IBM and European dwarfs

There were several companies who could have become an European IBM.
But why was there no European IBM?
Philips: secret agreements not to enter the computer industry.
Odhner (St. Petersburg 1873)
Machines were outnumbering IBM machines in 1917 in European offices.
Lots of Odhner copies all over Europe.
However, Odhner machines were old-fashioned because there were no new machines coming out of the factory.

To summarize, there was no European IBM because:
Companies signed agreements to not enter the computer industry (Philips).
Companies wanted to stay on the safe side of technology (Aatvidaberg Industrie).
Companies didn't innovate their machines (Odhner).

However, there were some European computer manufacturers that were influential. These are called the European dwarfs:
1949: Zuse KG with Z-line of computers
Begins production of Z4.
Zuse KG mainly focused on the engineering market.
It focused on doing calculations, not administration like IBM.
Zuse KG machines were not a huge success because it was a niche market within the German-speaking nations.
Zuse sold the company to Siemens AG in 1967.
New R&D was expensive, especially in Europe.
In the US, it was cheap to fire someone.
In Europe, it was difficult to fire someone once you had hired them (because of labor agreements).
1956: Electrologica (Dutch example)
Electrologica X1 produced in Amsterdam in 1956.

Expensive
Incredible success
Started mass-producing
Mainly mathematicians celebrated the Electrologica X1 as the pinnacle of logical computing and logical thinking

Electrologica X8 was produced in 1964.
Electrologica was building a general-purpose machine but it focused on the engineering market, not the administrative businesses.
Electrologica was bought by Philips in 1966.

There was something that the European companies had in favor: the existing anti US attitude.

Many companies would favor European machines instead of American machines.
However: IBM dominated European market anyways

1.3 Part 3: Innovation and Expansion

1.3.1 Chapter 7 : Real Time: Reaping the Whirlwind

A real-time system is one that responds to external messages in “real time,” which usually means within seconds, although it can be much faster.

1.3.1.1 PROJECT WHIRLWIND

MIT’s Project Whirlwind start = design an “aircraft trainer” (simulator)
MIT. Jay W. Forrester, was placed in charge of the project.
cost $8 million instead of the $200,000
Proj- ect Whirlwind established real-time computing and “Route 128”
Forrester retained overall administrative and technical leadership, Robert R. Everett day- to-day.
ENIAC and EDVAC and Perry Crawford. Provided inspo for Forrester
problem= reliable storage technol-ogy. Forrester tried the electrostatic storage tube = meh
From late 1947 Forrester scrutiny from the ONR, Office of Naval Research his new funders

1.3.1.2 THE SAGE DEFENSE SYSTEM

George E. Valley, creates the Air Defense System Engineering Committee (ADSEC),
1950, Valley seeking computer technology
George Valley got ONR, and Air Force money for project
A few months after, Whirlwind -> larger project, Project Lincoln
1951 the Whirlwind computer was an operational system. Problem = storage
The solution to this problem: 1951 when Bill Papian and prototype core- memory system.
Gradually the technology was transferred to IBM, which transformed the prototype Whirlwind into the IBM AN/FSQ-7 computer.
The SAGE system consisted of a network of computers and centers
The real contribution of SAGE = technological spin-off to civilian computing.
For example, the development of printed circuits, core memories, and mass-storage devices
The use of CRT-based graphical displays created
SAGE system provided a training ground for software engineering talent:

1.3.1.3 SABRE: A REVOLUTION IN AIRLINE RESERVATIONS

first major civilian real-time project = SABRE airline reservations system for American Airlines.
First manual then system, called the Reservisor, both unideal
By 1953 American Airlines had reached a crisis.
Perry Crawford explore the development of a computerized real-time reservations system
it would be some years before cost-effective technology became available.Thus an impressive new punched-card system and an improved Reservisor = “technology intercept” strategy.
The SABRE system was fully operational in 1964,
By the early 1970s all the major carriers possessed reliable real-time systems.

1.3.1.4 THE VISA SYSTEM AND AUTOMATED TELLER MACHINES

Predecessor = Diners Club,
Before ATMs, banks developed machines that dispensed cash
1960s. Example Barclays, the first British bank to launch a computer center (in 1961), De La Rue Automatic Cash System (DACS) was put into operation by Barclays in North London.
Credit cards faced the same challenges as the check system: authorization, clearing, and settlement.
In the late 1950s a number of banks began to experiment with charge cards. Bank of America, BankAmericard “credit card” in 1958
Other banks make National BankAmericard Incorporated (NBI) in 1970 which became Visa in 1976 with head = Dee Hock
NBI/Visa set out to build two computer sys- tems, BASE I and BASE II—the former for authorization and the latter for clearing and settlement.
Plus processing centers had a custom-engineered minicomputer
NBI/Visa’s BASE II was America’s first nationwide automated clearinghouse.
1980s Visa incorporated magstripe technology and inexpensive point-of-sale terminals
in 1975. Visa Debit.
Automated Teller Machine (ATM) networks. Pre ATM updated in batch overnight.

1.3.1.5 THE UNIVERSAL PRODUCT CODE

The Universal Product Code (UPC) is the barcode
the su- permarket, which replaced traditional food stores
The most obvious area for productivity improvement in a supermarket was the checkout, new generation of point-of-sale terminals that offered the possibility of automatic data capture for pricing and stock control.
In August 1970 an “Ad Hoc Committee of the Grocery Industry” was set up,
agree upon a product code of ten digits—five digits representing the manufac- turer and five representing the product.
European Article Numbering system somewhat later, adopted a six-plus-six code.)
In the spring of 1973 the barcode system with which we are now all familiar was formally adopted. This system was largely developed by IBM, and it was selected simply for being the cheapest and the most trouble- free option. The scanning technology was inexpensive, scanners manufactured by IBM and NCR.
For the food industry itself, there were negative impacts of bar coding, much as the rise of supermarkets in the 1930s spelled an end to the owner-operated grocery store.

1.3.2 **Chapter 8: Software**

1.3.2.1 **FIRST STEPS IN PROGRAMMING**

The discovery of debugging Maurice Wilkes, thoughts away from hardware and toward the programming problem.
John von Neumann and Herman Goldstine assumed that the conversion of the human-readable program into binary would be done by the programmer
Wilkes, however, realized could be done by the computer itself, Wilkes turned over the programming problem to David Wheeler , wrote a very small program, Initial Orders, that would read in a program that was punched in symbolic form on teleprinter tape, convert it to binary, and place it in the memory of the computer ready for execution.
The Cambridge group decided that the best way of reducing errors in programs would be to develop a “subroutine library”
In 1951 the Cambridge group put all their programming ideas into a textbook entitled The Preparation of Programs for an Electronic Digital Computer,

1.3.2.2 **FLOWCHARTS AND PROGRAMMING LANGUAGES**

flowchart. von Neumann began applying them to computer programs in the 1950s.
American research and computer manufacturers experiment with “automatic programming”
conservative culture of 1950s programmers didnt want to change
Grace Murray Hopper, the first programmer for the Harvard Mark I in 1943, then with UNIVAC.
proselytizing the virtues of automatic programming
At UNIVAC, for example, Hopper made the A-0 com- piler then successor = B-0 (or Flow- Matic) specif- ically for business applications, but it, too, was slow and inefficient.
IBM, John Backus. The system he produced was called the Formula Translator—or FORTRAN the first really successful “program- ming language,” used in model 704 computer.
new version of the language and compiler called FORTRAN II. Others example MAD
There was also an attempt to develop an international language, called ALGOL,
although ALGOL survived for a few years in Europe, was soon ousted by FORTRAN there.
While FORTRAN was becoming the standard for scientific applications, another language was emerging for business applications, COBOL—an acronym for the COmmon Business Oriented Language standard forced by the action of the US government. Committee on Data Systems and Languages (CODASYL) - COBOL 60
Hopper played an active role in promoting COBOL. design was influenced by Flow- Matic

1.3.2.3 **SOFTWARE CONTRACTORS**

the late 1950s computer manufacturers develop applications for specific industries(free)
free exchange of pro- grams within cooperative user groups such as SHARE
software contracting: firms that wrote programs to order for computer users.
One market = was for very large programs. The second market was for “custom” programs .
1956, the System Development Corporation (SDC), which became America’s first major software contractor.
In the late 1950s and early 1960s several other major defense contractors and aerospace companies—such as TRW, entered the large-scale software contracting business.
large software systems by the computer manufacturers themselves.
example, NCR developed retailing applications and Burroughs targeted the banks.
Probably the first small-time software contractor was the Computer Usage Company (CUC),
The first half of the 1960s was a boom period for software contractors.

The specialist software contractors joined by computer services & consulting firms ex. EDS,

1.3.2.4 THE SOFTWARE CRISIS

Not enough experienced programmers to do the work.
explanation software crisis was that the power and size of computers were growing much faster than the capability of software de- signers to exploit them.
If the 1960s was the decade of the software debacle, the biggest debacle of all was IBM's operating system OS/360. The "operating system" leader of project Frederick P. Brooks Jr.
OS/360 was the biggest and most complex program , system was "painfully sluggish"
OS/360 finally limped into the outside world in mid-1967, contained dozens of errors

1.3.2.5 SOFTWARE AS AN INDUSTRIAL DISCIPLINE

Explanation for software crisis "computer people" = different breed of tech expert, ie weirdos difficult nerds
into the 1960s there were no formal academic programs in programming:
It was also not clear what skills and abilities made for a good programmer:
growing demand for programmers
one solution was to reduce the level of expertise required of programmers.
COBOL = the desire to create computer code that could be read, understood, and perhaps even written by corporate managers.
Solution? = to develop a more rigorous academic approach to software development.
The first professional society for computer specialists was founded in 1946 as the Association for Computing Machinery (ACM).
The ACM was instrumental in establishing university-level departments of computer science
Vocational trade schools in computer programming emerged as an alternative to university
The most visible and influential response to the software crisis was the "software engineering" movement. Garmisch, 1968 NATO conference "software factory" "structured design methodology."
The structured-programming concept was also embodied in new computer languages, such as Pascal, invented in 1971,
Also solution = use of formal methods, good for small stuff not big
A much more widely adopted concept. The development model viewed the software writing process not as a once-and-for-all construction project, but as a more organic process,

1.3.2.6 SOFTWARE PRODUCTS

This situation created an opportunity for existing software contractors to develop software products—packaged programs,
One of the first packaged software products 1965 the software contractor Applied Data Research (ADR) released Autoflow,
The development of the software-products industry was accelerated by IBM's unbundling decision of December 1968,
IBM, however, was at this time under investigation by the antitrust division of the Department of Justice antitrust suit
software from free good to tradable commodity.
Perhaps the most spectacular beneficiary of the new environment for software products was Informatics, the developer of the top-selling Mark IV file management system—one of the first database products

By the mid-1970s all the existing computer manufacturers were important members of the software-products industry. IBM advantages: preexisting users and the ability to lease its software cheap
The most successful software-products supplier of the late 1970s and 1980s was Computer Associates.By the 1990s, the old-line software-products industry was being eclipsed by Microsoft and the other leading personal-computer software companies.

1.3.3 **Lecture 4**

Agendas
Things that people want/ try to achieve.For example:
- Selling machines: They wanted to be the top seller
- Academic discipline: People wanted an academic discipline in computer science/ electronics
- Thinking machines

Within in the academic world there were multiple different agenda's:
- Cybernetics
- Logic (Turing)
- Sharing Information
- Ordering sorting, processing
- Calculations

Real-time computing was also an agenda in the 1960's. It was a desire for engineers to achieve.
- Nowadays it cannot be seen as an agenda because we know that it exists.

Real time computing - the European story
Banks wanted to be able to offer their customers real-time service,so they invented the Cash Dispenser in 1967.
- It was seen as the 'robot cashier'
  - The machine could give you cash when banks were closed.
- At first ,checks were used to keep track of transfers and they were processed in batch mode. This resulted in the customers not having real-time information about how much money they had.

US had an ideal of the 'cashless society'. This was hardly present in Europe in the 1960's In 1969 the Cash Dispenser was replaced by the ATM.

The Barclays bank presented themselves as a modem bank.
- They prepared themselves for D-Day (Feb 15 1971), which stood for 'Decimal Day'.Decimals would be added to prices in England.

They wanted to work together with Burroughs for real-time machines.
  - Burroughs would sell a combination of the B5500 and TC500, the 88500 which would produce the most modern machine that Barclays needed.
  - The response time of the machine ended up being too slow.

By the end of 1968. Barclays merged with Martins, another bank with many offices.
The production of the 88500 was ended in 1969
  - This resulted in Barclays using batch-computing again
  - Instead of calling it 'real-time' they called in 'on-time'

The UPC code emerged 1n 1970 1971 in the US
- Motivated the real-time computing
- For start ups, it became more difficult to enter the food market

In Europe, they invented the European Article Number Code (1973-1974)
- It is known nowadays as lhe IPC
o The computing codes were made by IBM, but the scanners were sold by dwarf companies

Programming: ALGOL 60
Programs were initially created using wired machines or punched card machines
- In project Whirlwind the idea of 'autocoding' emerged to be able to set up a program in a more efficient way
- Other programming languages were made starting from flowcharts:
- FORTRAN. made by IBM
- COBOL. made by US government

European agenda was different:
- ALGOL
- The aim was to make a universal, elegant programming language
- Problems they wanted to solve with ALGOL:
  - Making existing programs run on new machines
  - Increase the ease of programming
- The US found the language too academic, not flexible and difficult to learn

software crisis
In the beginning of the 60's,there was software-industry rise
- The word 'software' was a confusing term in the 60's
o Companies that produced code never referred to software,they called it 'code' or 'program'
o Whereas external companies who tested programs called it a 'software project'

In 1968, IBM unbundled hardware and software
- Software became soft (something untouchable) and ware (something you could sell)
- Europeans did not like the idea of software being something you should pay for It was previously always included for free

Eventually, the software crisis happened.
- There were problems with growing amounts or code
- IBM had OS/360 drama

During the Garmish-Partenkirchen meeting in October 1968, there were some solutions suggested for the software cnsis:
- Making a universal programming language
- Using structured programming

1.3.4 Chapter 9 : New Modes of Computing

The personal computer grew out of time-sharing, the BASIC programming language, Unix, minicomputers, and new microelectronic devices.

1.3.4.1 TIME-SHARING

The first time-sharing computer system originated at MIT in 1961,
Predecessor batch-processing operating regimens.
The computing center at MIT was the first to implement a time-sharing sys- tem, in a project led by Robert Fano and Fernando Corbató. A demonstration version of MIT's CTSS—the Compatible Time-Sharing System—was shown in November 1961.

1.3.4.2 DARTMOUTH BASIC

Dartmouth Time-Sharing System (DTSS). intended for use by a much broader spectrum of users. by John Kemeny and Thomas E. Kurtz with General Electric
General Electric - hardware, software - Kemeny and Kurtz would have to develop their own. developed operating system + simple programming language called BASIC 1964.
BASIC—the Beginners All-purpose Symbolic Instruction Code
Kemeny, Kurtz, students began to write a BASIC translator and the operating software
Dartmouth BASIC became available in the spring of 1964,
Every computer manufacturer had to provide a BASIC translator for educational market.
BASIC's much more important cultural significance in establishing a user-friendly programming.

1.3.4.3 J.C.R. LICKLIDER AND THE ADVANCED RESEARCH PROJECTS AGENCY

time-sharing -> mainstream by Advanced Research Projects Agency (ARPA) 1962
MIT psychologist and computer scientist, J.C.R. Licklider. Man computer symbiosis dude
Lead to Information Processing Techniques Office (IPTO)
Project MAC— allowed users to do simple calculations, write programs, and run programs written by others; it could also be used to prepare and edit documents, paving the way for word-processing systems.
Successor = This system would be known as Multics (for Multiplexed Information and Computing Service).
To develop the operating software for the new system, MIT and General Electric were joined by Bell Labs. Having been pushed aside by Project MAC, IBM get into the time-sharing
August 1966 it announced a new member of the System/360 family, the time-sharing model 67.

1.3.4.4 THE COMPUTER UTILITY

time-sharing was broadened into the "computer utility" concept.
The idea of a computer utility from MIT around 1964 as an analogy to the power utilities.
computer users should be served by giant centralized mainframes from which they could draw computing power, instead of owning individual computers.
computer utility idea's drive = Grosch's Law = power of a computer varied as the square of its price. democratization of computing.
IBM was offering a service known as QUICKTRAN for simple calculations
The demise of the time-sharing no computer utility: the software crisis + lower hardware prices.
In 1970 General Electric decided to withdraw its model 645 computer from the market with the later it pulled out of the mainframe computer business altogether.

1.3.4.5 UNIX

spin-off Multics: the Unix operating system, developed at Bell Labs during 1969–1974.
small-is-beautiful approach to soft- ware
Unix was developed by two of Bell Labs' programmers, Ken Thompson and Dennis M. Ritchie,
With a PDP-7 manufactured by the Digital Equipment Corporation (PDP stands for "pro-grammed data processor").
By early 1970 Unix was working to its creators' satisfaction
Thompson and Ritchie had a hard time convincing their colleagues of its merits.
The system was completely rewritten in a language called "C," which was designed especially for the purpose by Ritchie. This was a "systems implementation language,"
The use of C made Unix "portable," so that it could be implemented on any computer system
A major change in the Unix world occurred in 1977, when it began to grow organically:
Unix was on the path to becoming the standard operating system for the 1980s
In the late 1990s Unix took on yet another mantle as the Linux operating system.

1.3.4.6 THE MICROELECTRONICS REVOLUTION

Hamilton Pulsar digital watch.cutting edge, a luxury item
Then digital watch became cheap with the cost of components going down
planar process for manufacturing integrated circuits, developed at Fairchild Semiconductor and required a substantial initial investment, but afterl cost of production dropped rapidly.
In order to continue to profit, semiconductor firms had to create new and ever-increasing demand for their products.ex. video game console

1.3.4.7 FROM MIT TO MINICOMPUTERS

Minicomputer is not scaled down mainframe
Minicomputer = Digital Equipment Corporation (DEC), by Kenneth Olsen and Harlan Anderson to en- ter the mainframe business needed:CPU, peripherals, software, sales force.
DEC announced its first computer, the PDP-1, in 1960.
Olsen aimed the PDP-1 at the science and engineering market. No peripherals/software=cheap
PDP-8 minicomputer in 1965. The PDP-8 w/ integrated circuits(one of the first)
The PDP-8 was an instant success, ->colleges and research laboratories,
Stephen Gray founded the Amateur Computer Society (ACS) in May 1966 in order to exchange information with other hobbyists about building computers.
By 1969, DEC had been joined by other minicomputer manufacturers, such as Data General
By 1970 it was the world's third-largest computer manufacturer, after IBM and the UNIVAC Division of Sperry Rand.

1.3.4.8 SEMICONDUCTORS AND SILICON VALLEY

For the first several decades of its existence, the geographical center of the computer industry was the East Coast with IBM and MIT
Beginning in the mid-1950s, however, a remarkable shift westward
the 1960s it was the Silicon Valley
The migration westward began with William Shockley, Shockley Semiconductor Laboratories, brought Robert Noyce,
By 1957 Shockley Eight = Robert Noyce and 7 others left cause WS an asshole
-> Fairchild Semiconductor In 1959 Fairchild cofounder Jean Hoerni developed a novel method for manufacturing transistors called the planar process,
Noyce, then head of research at Fairchild Semiconductor, extended the planar process,

1.3.4.9 THE FAIRCHILDREN

Famous “Fairchildren” = Intel, by Robert Noyce and Gordon Moore recruited from Fairchild
new breed of Silicon Valley start-ups = fast, flexible, and meritocratic.
During the 1960s the pace of semiconductor innovation was furious. There were two main reasons for this. The first was technological: the planar process,
By 1970 it was possible to make LSI chips (for Large Scale Integration) during the early 1970s they decimated the core-memory industry.
The rapidly increasing density of semiconductor chips was first commented upon by Gordon Moore “increased at a rate of roughly a factor of two per year” - “Moore’s Law,”
A second reason: nature of the early market.
the guaranteed market for semiconductors provided by the military, combined with the forced circulation of knowledge encouraged competition, innovation, and a relentless pursuit of incremen- tal product improvement.

1.3.4.10 CHIPS IN CONSUMER PRODUCTS

The “upside-down economics” An excellent example of this is their expansion into the calculator in- dustry. The first hand-held calculators were introduced around 1971 at a price of $100; by 1975 this had fallen to under $5. Of these new users of high-density integrated circuits, perhaps the most signifi- cant was the video game industry. Atari was founded in 1971 by Nolan Bushnell
Its first product was an electronic table-tennis game called Pong Whereas other firms were devastated by the collapse in chip prices, Bushnell saw an opportunity to move Atari from low-volume production for arcades to high- volume production for the domestic market,Because the value of a video game system was largely in the soft- ware (the games themselves) the intense price cutting less effect
The two technologies were closely related, with many microcomputers (as they were then known) being purchased primarily to play games, and many video game man- ufacturers (including Atari) also manufacturing microcomputers.

1.4 Part 4: Getting Personal

1.4.1 Chapter 10:The Shaping of the Personal Computer

1.4.1.1 RADIO DAYS vs. Personal Computer

Three key points: First, radio came from a new enabling technology whose long-term impor- tance was initially unrecognized
Second, a crucial set of actors in this transformation were the radio amateurs.
Finally, once radio broadcasting was established, it was quickly dominated by a few giant firms

1.4.1.2 MICROPROCESSORS

1969–1971 in the semiconductor firm Intel. by Intel engineer Ted Hoff,
Originally for Japanese calculator manufacturer Busicom
The new calculator chip, known as the 4004,

Successor - The 8080, which became the basis for several personal-computer designs, appeared in April 1974. By this time other semiconductor manufacturers were starting to produce their own microprocessors—such as the Motorola 6800,

1.4.1.3 COMPUTER HOBBYISTS AND "COMPUTER LIBERATION"

Most hobbyists had some professional competence. Want home computer
The allure of the microprocessor = reduce the price = home computer
The computer hobbyist was primarily interested in tinkering with computer hardware;
the advocates of "computer liberation., small movement: a widely held desire to bring computing to ordinary people. particularly strong in California, and this perhaps explains why the personal computer was developed in California rather than, say, around Route 128.
strong anti-establishment culture, tech not freely available; it was mostly rigidly controlled in government bureaucracies or pri- vate corporations.
Some from student-led New Left, or "New Communalists." like Stewart Brand
"The Whole Earth Catalog". inspirational to Steve Jobs,
Ted Nelson, idea called hypertext, self-published books Computer Lib and Dream Machines,.
He influenced mainly the young, predominantly male, local Californian technical community.

1.4.1.4 THE ALTAIR 8800

In January 1975 the first microprocessor-based computer, the Altair 8800, for hobbyists
In every other sense the Altair 8800 was a traditional minicomputer. Self assembled
By Albuquerque, New Mexico, Micro Instrumentation Telemetry Systems (MITS).
The limitations of Altair 8800 = opportunity for small-time entrepreneurs to develop "add-on"
A few other entrepreneurs developed software for the Altair 8800. ex. Bill Gates
Gates programmed in BASIC with his buddy Paul Allen -> Traf-O-Data.(firm)
develop a BASIC programming system for the new machine.
Gates and Allen formed a partnership they named Micro-Soft, they delivered a BASIC programming system to MITS in February 1975.
1975, "Homebrew Computer Club" Silicon Valley. Besides acting as a swap shop for computer components and programming tips, computer- hobbyist and computer-liberation meld.
During the first quarter of 1975, MITS first "worldwide" conference. At this conference Gates:
does some capitalism, no free software :(, much anger
new computer magazines ex., such as Byte, -> selling computers by mail order,
Mail order -> computer stores such as the Byte Shop and ComputerLand,
1975, new models produced by firms such as North Star

1.4.1.5 THE RISE OF APPLE COMPUTER

Apple was founded by two young computer hobbyists, Stephen Wozniak and Steve Jobs.
First they went into business making "blue boxes"—
attending the Homebrew Computer Club in early 1975.
Wozniak realized new microprocessor technology and, computer based on the MOS Technology 6502 chip. He and Jobs called it the "Apple,"
To be a success as a product, the microcomputer = self-contained unit
Most important, the machine would need software to appeal to anyone other than an enthusiast.

1.4.1.6 SOFTWARE: MAKING PERSONAL COMPUTERS USEFUL

During 1977 : Apple, Commodore Business Machines, and Tandy,

West Coast Computer Faire in April 1977, first two machines for the mass consumer, the Apple II and the Commodore PET, were launched
the Commodore PET similar to Apple II, not so much a computer as a calculator writ large.
the PET was a closed system, with no potential for add- ons
Later Tandy, entered the market, when it announced its TRS-80 by Radio Shack,
Jobs's envisioned the machine also being used as an appliance in the home
the barriers to entry into personal-computer software were low = more firms(they die alot)
The most popular products included Microsoft's BASIC programming language
This software was usually bundled with the machine
market for "applications" soft- ware took off.
There were three main markets for applications software: games, education, and business.
The market of packaged software for business applications developed between 1978 and 1980, when three generic applications enabled the personal computer to become an effective business machine: the spreadsheet, the word processor, and the database.
The first application VisiCalc spreadsheet. By Daniel Brisklin = personal-computer revolution
Word processing on personal computers did not develop until about 1980.
availability of low-cost printers from Japanese manufacturers,helped word-processing market.
The first successful firm to produce word-processing software was MicroPro 1978
Word- Master in 1978. -> WordStar

1.4.1.7 THE IBM PC AND THE PC PLATFORM

personal computer = clearly defined as a business machine in 1980, IBM reacted proposal that IBM should enter the personal-computer business came from William C. Lowe:
outsourcing all the components it did not already have in production, including software
The development of the personal computer would be known internally as Project Chess.
IBM could make use of the second generation of microprocessors
IBM did not have the skills to develop software for personal computers. = deal with Microsoft software from a local software firm, Seattle Computer Products,
1980 the prototype personal computer, known internally as the Acorn, was complete;
PC sold by Sears and ComputerLand, Lots of marketing and press = IBM had legitimated the personal computer.
During 1982–1983 the IBM Personal Computer became an industry standard.
clone manufacturers was Houston-based Compaq, which produced its first machine in 1982.

1.4.1.8 DELL COMPUTER AND PROCESS INNOVATIONS

Michael Dell, dorm room was not merely the location of early-planning and prototype design but also the initial site of product assembly.
Dell began a business building upgraded PCs and add- on components in his dorm room to sell to local businesses. -> PC's Limited.
1985 Dell hired engineer Jay Bell to create Dell's first PC,
sold computers directly by mail order =cheaper
A key to his success was mass customization—
The company became Dell Computer Corporation in 1988 to take advantage of the name recognition of its famed founder.
By 1986 the company had gained extensive free publicity from building the world's fastest PC, direct sales solidified relationships with customers to achieve repeat business
Dell Computer Corporation, Just-in-time basis for components first to do in computer trade, no extra inventory

Dell’s direct sales model also brought imitators—most prominently Gateway Computer Corporation,
By far the greatest beneficiaries of the mas- sive growth of personal computers were Intel and Microsoft.
Almost all of the companies that resisted the switch to the IBM standard soon went out of business
exception was Apple Computer, by making cheaper hardware but by mak- ing better software.

1.4.2 **Lecture 5 : Dream Machines and Modes of Computing**

Xanadu-project (1960) first hypertext project
European valleys not like silicon valleys tech valleys did exist (mimic Silicon) Europe: academics and high tech companies were interested in time sharing facilities EXAMPLE: campus Twente polytechnic(Enschede, 1964) but no one wanted to live there, success in 90s h
Europe: 1960s academics and high tech companies but no interest for computers in youth movements
1973 Amateur Computer Club(ACC) by Mike Lord (Essex, 1973) examples of the more rare amateur or youth groups interest in computers in Europe
Europe: hippies in Europe were political but not computer people, and vice versa
Clubs Appropriating the machine Altai 8800 1975
Amateur Computer Society first personal computer club 1966
States: Homebrew Computer Club, Europe: Hobby Computer Club 1977 = HCC
In the course of the 1980s the clubs changed becoming computer organisations. HCC guarding the interests of users
Squatter movement in the netherlands and europe 1980s. Squatter movement = came about among students and others protesting against the lack of housing and claim empty houses
Squatter movement was keen on using machines and networking
Chaos Computer Club 1984 Hamburg - the combo of political human rights and computers they were also hackers and were for general access to computers and general education. Freedom of data. And they hacked deutsche bank. Anarchists :) Steffen Wernery dude important
By end of 80s hacking bad illegal etc. before a fun hobby and something done in clubs
Political activism in Europe mid 1980s onwards in contrast to 1970s in the states
Europe a bit behind in information sciences in the netherlands especially the netherlands lacked IS academic setting while it existed in US, France, UK
SSAA: founded 1958 and had a journal and provided info seminars
Max Euwe member SSAA he was a UNIVAC ambassador. embodied IS in netherlands
SSAA and HCC were not collaborators
But a dutch government funded project and BBC made Teleac television courses about microprocessors and programming ie educational programs late 1970s
Chriet Titulaer promoted tech and science made those tech television courses
Computers in education as an instance of appropriation: Computer for motivation to learn math, familiarization
In education on three levels: curriculum, in thinking about education and how to educate people, and in the educational process in classroom in textbooks, Computers seen 1960s in binary arithmetic classes, flow charts (Example Grace Hopper educator)
1954 Skinner machines starting to make appearances in classrooms
Programmed instructions influenced by computing technology and vice versa
Positive influence on general trust and familiarity in computing technology
1970s computing facilities were rare. In the 1980s computing facilities in classrooms either funded by governments or companies

Then there is a change in movie culture 1970-2000 Going digital movies and themes become more digital like Blade Runner plus specific effects propagated by computers. Transition away from physical stunts

1.4.3 **Chapter 11: Broadening the Appeal**

1.4.3.1 **MATURING OF THE PERSONAL-COMPUTER SOFTWARE INDUSTRY**

The personal-computer software industry developed in two phases.
The first phase, = gold-rush era, lasted from about 1975 to 1982; barriers = extremely low,
The second phase, which began about 1983, following the standardization IBM-compatible PC, = consolidation a small number of (American) firms emerged as global players.
New industry was its almost total disconnection from the existing software-products industry, . capabilities of the existing software firms no = developing programs for first personal computers;
cultural reasons: the personal-computer software companies sold their products through mail order and retail outlets, one is for the people, one for the businesses
Despite the large number of entrants -> market leaders. These included the VisiCalc spreadsheet, the WordStar word processor, and the dBase database by 1983
a critical success factor was marketing.
three significant barriers: improving performance of personal computers, know-how, access to distribution channels.= importance of a single “hit” product
Example VisiCorp publisher of VisiCalc wiped out by competing product 1985
Lotus Development Corporation, Mitch Kapor, with Lotus 1-2-3, bye VisiCalc.
Lotus went hard on publicity and marketing and became market leader
Microsoft illustrates the value of an income stream MS-DOS from IBM
Microsoft diversify without having to rely on external sources of capital.
Microsoft unsuccessful products Example: Microsoft’s first application was a spreadsheet called Multiplan, but it was eclipsed by Lotus 1-2-3
November 1983 launch of Word: Word was initially not a successful product and had a negligible impact on the market leader, WordStar.

1.4.3.2 **GRAPHICAL USER INTERFACE**

For the personal computer had to become more “user-friendly.” graphical user interface.
Until Macintosh, personal computers communicated with the user through a disk operating system, or DOS. example MS-DOS. Based on Unix operating sys- tem.
difficult and irritating. through a “command line inter- face,
The graphical user interface (GUI, pronounced “gooey”) natural and intuitive
The GUI = WIMP interface: Windows, Icons, Mouse, and Pull-down menus.
“Almost all the ideas by ARPA’s Information Processing Techniques Office in the 1960s:
The Human Factors Research Center with Doug Engelbart, later regarded as the doyen of human-computer in- teraction. best-known invention is the mouse. 1965
Engelbart’s work “ inspo for graphical user interface at the Xerox Corporation.
also David Evans and Ivan Sutherland, focus computer graphics. With grad student Alan Kay
Kay made a system for The Dynabook was to be a personal-information system, the size of a notebook, that would replace ordinary printed media.
existing in the 1960s what took so long?lack of a compact and cost-effective technology.
in 1969 Xerox researches “the office of the future.”
Xerox recruited Robert Taylor,. Taylor fan of Licklider’s “man-computer symbiosis” vision,
= “Alto” desktop computers. began in 1973, before the advent of personal computers

Xerox decided to produce a commercial version of the Alto, called the Xerox Star 1981
too expensive not enough marketing

1.4.3.3 STEVE JOBS AND THE MACINTOSH

1979 Steve Jobs saw Alto -> The Lisa launched in May 1983, expensive
beyond the budget of personal-computer users and even beyond that of the corporate business-machine market. Lisa was a resounding commercial failure.
Idea: Jef Raskin, then manager of advanced systems at Apple.
Macintosh as an “information appliance,” which would be a simple, friendly computer that the user would simply plug in and use.
the project was appropriated by Jobs. Raskin left Apple in the summer of 1982.
design effort in the Macintosh went into obtaining Lisa-like performance at a much lower cost.
Consumer marketing = ad in Super Bowl 1984:
Macintosh sales disappointing.
October 1983 IBM had launched a new low-cost machine, the PC Jr., unsuccessful
Microsoft to develop a range of technologically sophisticated applications for Macintosh, largely insulated from the much more competitive IBM-compatible market.

1.4.3.4 MICROSOFT’S WINDOWS

1981, GUI-based operating system for the PC—including Microsoft, Digital Research, and IBM itself for the IBM-compatible PC
very demanding for two reasons.
First, the PC underpowered for GUI. Second, replace or layer with MS-DOS
Digital Research did develop a PC operating system called CP/M 86, died on vine -> GEM operating system. lacked the capability of a full-scale operating system.
IBM’s TopView 1984 too slow
Microsoft Windows last to appear first called Interface Manager,
version 1 of Windows was released in October 1985
heavily based on the Macintosh user interface— Microsoft signed a licensing agreement with Apple to copy the visual characteristics of the Macintosh.
“unbearably slow.”
late 1980s, with the Intel 386 and 486 more power
In April 1987 IBM and Microsoft announced their intention to jointly de- velop a new operating system, OS/2 replacement for MS-DOS.
1987 Microsoft announced a new release of Windows, version 2.0. Looked just like the Macintosh. Apple filed a lawsuit Suit eventually dismissed
Success + Microsoft’s marketing was much better.
On 22 May 1990 Windows 3.0

1.4.3.5 CD-ROMS AND ENCYCLOPEDIAS

the speed and capacity of networks was simply too limited for supplying large volumes of information. Because modem
CD-ROM disc = large volumes of information were shipped to home computers. by Sony and Philips
New market - Gary Kildall and Bill Gates, played critical roles in establishing a market for CD-ROM media.
Kildall started a new CD-ROM publish- ing operation as a separate company
Bill Gates promoted it
the cost of CD- ROM drives had fallen = mass market for CD-ROM media.

Ex. Grandma and Me

1.4.3.6 THE RISE AND FALL OF VIDEOTEX

in the United States, Europe, and Japan, vibrant information networks had already been established for businesses and consumers by the 1980s.
Whereas in the United States such networks were largely developed by the private sector, elsewhere gov- ernment sponsorship played a much greater role. The dominant technology of these subsidized networks was known as videotex.
Videotex evolved from a technology developed in the mid-1960s by RCA known as teletext,
De- velopment of international standards and a videotex global network were hampered by politics and national pride as well as by the fact that each country had developed its own display standards.
France. The French system, Minitel, first came. France's eventual expansion of telephone service and infrastructure
rapidly rising costs of printing and distributing phone books, thereby providing the impetus for Minitel to take off.
French government distributed units and developed an online telephone/address directory that people could use for free.
"network effects": more Minitel users = more services = more users. A further contribution to Minitel's financial success sex-chat services.
the existence of Minitel inhibited France's mass adoption of the Internet until the early 2000s,

1.4.3.7 CONSUMER NETWORKS IN THE UNITED STATES

leading consumer network throughout the 1980s was CompuServe. an old-line time-sharing service
Jeff Wilkins sell this spare capacity to computer hobbyists. he created a service called MicroNet.
arrangement with the Tandy Corporation to distribute a $39.95 "starter kit"
PCs moved into the workplace, CompuServe also tapped the business market, offering "Executive Services"
CompuServe began to attract some major competitors. The two most important were The Source and Prodigy.
The Source was a network set up by the Reader's Digest Association and the Control Data Corporation, while Prodigy was created by Sears and IBM.
CompuServe's relative success benefited from network effects.
mid- 1980s: America On-Line (AOL). By making the service free to try out, it encouraged hundreds of thou- sands of consumers to do just that. And because the service was user-friendly, many of them stayed on as paying customers.

1.4.4 Chapter 12 : The Internet

1.4.4.1 I. From the World Brain to the World Wide Web

1.4.4.1.1 FROM THE ENCYCLOPEDIA TO THE MEMEX

H. G. Wells, convinced that narrow specialization—causing the world to descend into barbarism
project for a World Encyclopedia - "network" for the World Brain
Project World Brain - reappears by Vannevar Bush,
Bush envisaged a personal-information machine that he called the memex:
new technology of comput- ers would be instrumental in realizing the memex.

1.4.4.1.2 THE ARPANET

In 1963, when the first ARPA-sponsored time-shared computer systems became operational, Licklider set into action a program he privately called his Intergalactic Computer Network, publicly known as the Arpanet.
the users of each computer would be able to use the fa- cilities of any other computer on the network;
Larry Roberts, to head up the project.
three major technical problems
The first problem was how to physically connect all the time-sharing systems together
The second problem was how to make economic use of the expensive high-speed communications lines connecting com- puters.
The third problem Roberts faced was how to link together all the computer sys- tems, which came from different manufacturers
Unknown to Roberts, a solution to the first two problems had already been in- vented. Known as "store-and-forward packet switching,"
Instead of having every computer connected to every other, store- and-forward technology would be used to route messages through the network; there would be a single "backbone"
The computers that acted as the switching centers—called nodes in the Arpanet
The final problem = solution: the interface message processor (IMP).
, a separate inexpensive minicomputer, the IMP,
The software on each computer system— called a host—would need only a relatively simple modification to collect and deliver information between itself and the IMP.
The participation of so many graduate students in developing Arpanet, created a distinctive and somewhat anarchic culture in the network com- munity. accounted for the unstruc- tured and anarchic state of the early Internet.

1.4.4.1.3 THE POPULARITY OF E-MAIL

communicating through electronic mail that attracted users.
Electronic mail had never been an important motivation for Arpanet.
Electronic mail soon exceeded all other forms of network traffic on Arpanet,
The demand for e-mail fa- cilities was also a major driving force for the first non-ARPA networks.
Ex. Usenet,
First, the time-sharing firms began to restructure themselves as network providers. "The Great Electronic Mail Shootout."
the more people there were on the networks, the more useful e-mail became.
e-mail eliminated the need for both parties having to synchronize their activities, freeing them from their desks.
Gradually an unwritten "netiquette" emerged, .
While some of the new computer networks of the 1970s were based on the technology developed in Arpanet, this was not true of all of them. Computer manufacturers, in particular, developed their own systems such as IBM's Systems Network Architecture (SNA)
Fortunately, IPTO idea, internetworking, was made concrete as the Internet. The job of IPTO at this stage was to establish "protocols" by which the networks could communicate.
The system that ARPA devised was known as the Transmission Control Protocol/Internet Pro- tocol, or TCP/IP—

1.4.4.1.4 THE WORLD WIDE WEB

late 1980s most professional computer users (though not home users) in the United States had access
Once this much broader community of users started to use the Internet, people began to exchange not just e-mail but whole documents.
soon there were millions of documents but no catalog and no way of finding what was useful
Numerous parallel developments were under way to create finding aids for in- formation on the Internet. One of the first systems, “archie,”
A more impressive system was the Wide Area Information Service (WAIS),
The most popular early finding aid was “gopher,”
Hypertext make it unneces- sary to locate documents in centralized directories. links would be stored in the documents themselves, and they would instantly whisk the reader to related documents. Like memex
The World Wide Web was invented by Tim Berners-Lee. Inspo from hypertext
Berners-Lee hobby program for a hypertext system he called Enquir
Berners-Lee has described the World Wide Web as “the marriage of hyper- text and the Internet.”
Berners-Lee and his Belgian collabora- tor Robert Cailliau
There were really two sides to the World Wide Web concept: the server side and the client side.
The server would deliver hypertext documents (later know as web pages) to a client computer,
By 1991 Berners-Lee and Cailliau WWW paper was turned down,
After that WWW took off
The new generation of browsers (such as the University of Helsinki’s Erwise and Viola at UC-Berkeley) provided not only the user-friendly point-and- click interface that people were starting to demand but also full-blown hypermedia features.
The best-known server program to evolve by this process was known as “apache,” a pun on “a patchy” server.
The Internet enabled collabora- tive software development (also referred to as the open-source movement) to flourish to an extent previously unknown.
Berners-Lee persuaded CERN to place web technology in the public domain, so that all could use it free of charge for all time.

1.4.4.2 II. The Web and Its Consequences

The Mosaic browser from the National Center for Supercomputer Applications (NCSA)
by Marc Andreessen,

1.4.4.2.1 BROWSER WARS

on 4 April 1994 the Mosaic Communications Corporation was incorporated to develop browsers and server software for the World Wide Web changed to Netscape Communications
In December 1994 Netscape shipped version 1.0 of its browser and complementary server software.
Microsoft, far and away the dominant force in personal com- puter software, was seemingly oblivious to the rise of the Internet. Its online service MSN was due to be launched at the same time as the Windows 95 operating sys- tem in August 1995.
Microsoft covered its bets by licensing the Mosaic software from Spyglass and including a browser dubbed Internet Explorer with Windows 95, but it was a lackluster effort.
Microsoft was not the only organization frozen in the headlights of the Internet juggernaut.
Consumers choice was between the mature, user-friendly, safe, and content-rich world of the consumer network or the wild frontier of the World Wide Web.

ISPs did not provide much in the way of content The ISP gave you a connection, but then you were on your own.
AOL licensed Microsoft's Internet Explorer for use in its access software. This gave its subscribers the best of both worlds—the advantages of the existing service plus a window into the World Wide Web. CompuServe did much the same, though it was eventually acquired by AOL in 1998.
Microsoft, on the other hand, decided to cut its losses on MSN.
In 1995 Netscape had seemed unstoppable.
During the next two years Microsoft and Netscape battled for browser suprem- acy,
version 4.0 of Microsoft's Internet Ex- plorer achieved technical parity with Netscape. Because Internet Explorer was bun- dled at no extra cost with the new Windows 98 operating system, it became the most commonly used browser on PCs
A version of Internet Explorer was also made available free of charge for the Macintosh computer.
Microsoft had fought hard, perhaps too hard, for its browser monopoly. Its alleged practices of tying, bundling, and coercion provoked the US Department of Justice into filing an antitrust suit in May 1998.

1.4.4.2.2 THE INTERNET LAND RUSH

commercial revolution; users were able to purchase the full range of goods and services that society could offer.
In 1990, however, the Internet was largely owned by agencies of the US Government, and the political establishment did not permit the use of publicly owned assets for private profit. Hence privatiza- tion of the Internet was an essential precursor for electronic commerce to flourish.
National Science Foundation had created another network, NSFnet. eventually becoming the "backbone" of the entire Internet.
By 1995, the commercially owned parts of the Internet far exceeded those owned by the government. On 30 April 1995, the old NSFnet backbone was shut down, ending altogether US government ownership of the Internet's infrastructure.
The explosive growth of the Internet was in large part due to its informal, de- centralized structure;
Rules in internet A good example of this is the domain-name system.
Yahoo! began in the days of the Mosaic browser as a simple listing service created by David Filo and Jerry Yang, Yahoo! was not without competition: Lycos,
profit via advertising was the obvious choice.
Larry Page and Sergey Brin, created their "PageRank" algorithm Thus was born their "search engine," Backrub, renamed Google shortly be- fore they launched the URL google.stanford.edu in September 1997.
In 1998 Page and Brin launched Google Inc. in a friend's Menlo Park garage.
when users clicked on these sponsored links, Google was paid a small sum. .
Another early web success was mail-order selling. Jeff Bezos's Amazon.com es- tablished many of web commerce's early practices (including the incorporation of com in the firm's name).
book- selling Launched in July 1995, Amazon.com would quickly become the leading online bookseller and, in time, the world's top online retailer.
dot-com crash of 2000.
Google successfully moved into e-mail services (Gmail), maps and satellite photos, Internet video (with its 2006 acquisition of YouTube), cloud computing, digitizing books, and other endeavors. More recently, Google has also been an important participant in open- source mobile platforms that are transforming computing.

1.4.4.2.3 GOING MOBILE

From shortly after the advent of personal computing, computers have become in- creasingly mobile.
in 1982, GRiD Systems Corporation launched the GRiD Compass 1101, arguably the first laptop computer.
Various models of commercial portable computers hit the market as the decade progressed, including the Tandy TRS-80 100, the first popular portable to use an LCD screen.
These portables, however, gave up much in screen size, processing power, memory, and compatibility to achieve degrees of mobility.
Smart- phones largely evolved from an earlier technology, Personal Digital Assistants (PDAs),
Apple's iOS and Google's Android. These two new platforms benefited greatly from third-party ap- plication software providers.
Psion, a UK software company formed in 1980, soon diversified into hardware and in 1986 introduced the Psion Organizer II, an operating system–based PDA
Silicon Valley–based Palm, Inc. stumbled along with handwriting-recognition PDAs in the early 1990s before achieving suc- cess with its Palm Pilot in 1996.
Though Palm's cultivation of a modest ecosystem of third-party applications developers foreshadowed what would become critical in the smartphone market, memory was too small for truly compelling applications.
Research in Motion (RIM), a Canadian specialist in paging, messaging, data capture, and modem equipment that launched the PDA "Blackberry" in 1999.
In 1985 Jobs formed NeXT, a computer platform development company focused on the educational and business markets.
Apple, Jobs envisioned the firm's expansion into consumer electronics. Apple launched the iPod, but Apple's player rapidly captivated consumers and gained market leadership.
the key to the iPod's immense success was the simultaneous launch of Apple's iTunes store.
The iPhone, launched in 2007, quickly overcame being late to the smartphone market with its easy-to-use touchscreen and its successful cultivation of third-party applications (apps) developers.
The iPhone App store, resembling the familiar iTunes store, dwarfed the of- ferings of competitors and drove demand for Apple phones that for years sold at a substantial price premium to other smartphones.
Meanwhile, in 2005, Google acquired Android, Inc., a producer of an open- source, Linux-based mobile operating system.
Apple and Android extended their offerings to include popular tablet computers (for Apple, the iPad), which segued the features of a laptop with the mobility of a smartphone.
During 2010–2012 Microsoft made a major foray into consumer smartphones and tablet computers.

1.4.4.2.4 WEB 2.0

facilitating and encouraging user-generated content
Wikipedia (combining the Hawaiian term wiki, meaning "quick," with encyclo- pedia) was launched by Jimmy Wales and Larry Sanger in 2001.
Companies and organizations, which in the past tended to maintain their own data locally, increasingly are contracting with specialists such as Amazon.com And other leaders in the rapidly growing field of cloud computing—using remote servers for various data and software ap- plications needs. This allows the staff of client firms and organizations to readily access and share data with colleagues, customers, suppliers, and other authorized users—and to benefit from suppliers' economies of scale and expertise in data storage and delivery.

1.4.4.2.5 SOCIAL NETWORKING: FACEBOOK AND TWITTER

Cloud computing is also at the heart of online social networking.
MySpace (formed in 2003).
overshadowed by industry-leading Facebook.
Mark Zuckerberg founded Facebook—then called Thefacebook
Subtle and non- intrusive advertising was introduced to help fund the operation.
Facebook (the new company name, with the do- main www.facebook.com, as of August 2005) validated all users. This prevented the problem of fake profiles that at times plagued Friendster (sometimes derisively called “fakester”) and MySpace
2006 fall, Facebook became available to all users aged thirteen or older and soon began to concentrate on international expansion
As with Facebook’s growth, its features were added gradually and methodi- cally, and keeping the site simple and user-friendly .
Facebook’s greatest vulnerabilities—users’ pri- vacy concerns.
Twitter is Facebook’s closest rival, by Jack Dorsey in 2006

1.4.4.2.6 POLITICS OF THE INTERNET

As usage of the Internet greatly broadened with the advent of the World Wide Web, many journalists, politicians, and others have presented it as a transformative technology of freedom and democracy. Organizations such as the Electronic Fron- tier Foundation, formed in 1990 by Lotus Development founder Mitch Kapor and Grateful Dead lyricist John Perry Barlow, have emerged to defend individual Inter- net rights—with some such advocates from the political left and many others from the libertarian right.
Example, Journalists quickly christened the 2009 protests in Iran—against alleged fraud in reelecting president Mahmoud Ahmadinejad—as the “Twitter Revolution.”
How much and what types of regulation should be established for the Internet are bitterly contested questions within and between nations.

1.4.5 **Lecture 6**

Personal computers: more appropriation
Software prices Chaos Computer Club unhappy, software should be price free
HCC stance was yea if its a fair price pay for it
Chaos Computer Club(1982) hacking is a political statement
1984 Members of the CCC Herwart Holland and Steffen Wernery: hack Bildschirmtext, hacktivism
In 1988 Wernery arrested for computer crimes with Deutsche Bank
Friendly hack but later in the 80s was considered a crime before world thought hacking was bad
CCC(bigger political hacking) vs HCC(restrained hacking)
Demoscenes of CCC: European phenomenon 1980s (demonstrating things) came on “diskmags”(1984)
Diskmags happen in the states in 90s but eclipsed soon by internet
Diskmags goes with the “Free software movement” rose 1983 from MIT (more about accessibility and transparency than price)
US-free says free access and accessibility whereas Europe focused more on free in price as well
1988 Open source software movement
Example “Heroes of the computer revolution” book in 1984

Europe “Burgerinformatica” 1983 EU trusts their government more and the governments increase education of computer and in the netherlands end 1980s 30% ofpeople owned a pc and by 1995 increased to 60%
Netherlands: Kees van Kooten, Wim de Bie(1986) ridicule for having a computer ownership and not using it properly or at all
De Digitale Stad(1994) governments and organisations that promote access to digital stuff - the digital city online unification
France: Minitel from 1978 to 2012. It was funded by the government
The Netherlands: Viditel(1980) network that exists and Teletekst(1980)
The Netherlands: national provider Datanet 1(PTT)

- Early phase: circuit-switching telexlines
- Starting phase 1978-1981, transition to packet switching on the basis of X25-protocol (network running)
- Launched through existing network of telex-lines
- Successful with journalists, customs and banks
- Use of own networks and interconnected designated lines(by PTT) was an alternative for large companies. PTT offered an “affordable” alternative
- Wasnt as successful as France and US

Rise of game industry

- Playing as a way of appropriating
- Chess algorithms by Alan Turing (1952) no computer to play it on so he played it himself and sometimes lost against his algorithm
- Idea born to teach computer how to play chess
- Nimbus(1960)
- Spacewar!(1970) game created in MIT
- 1972 first real tournament for game
- Slow rise of arcade halls
- Vectorbeam(1977) on a DEC PDP-architecture(arcade game thing)
- 1971 Oregon Trail text and story based(made by teacher), MECC(1974) The Minnesota Educational Computing Consortium (later Corporation),
- Game computers in arcade halls
- Pong(1972) by Atari was already there (no text based adventure games)
- PAcman(1980) by Namco, a game changer so popular
- Nintendo from Japan too Donkey Kong (1981)
- Gaming industry boomed in the 80s

Administration ends up as Information Science
Process Control as artificial intelligence
Science becomes Computer Science